

Supplementary Material

Sex Differences in the Association Between Insulin Resistance and Systemic Inflammation Below the HbA1c Prediabetes Threshold

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Table S1: Sensitivity of the HOMA-IR-by-sex interaction to exposure scale and upper-tail restriction

HOMA scale	Analytic restriction	<i>n</i>	Female slope	Male slope	Male–female	Interaction <i>p</i>
Raw	Full sample	3079	0.0629	0.0082	−0.0547	0.00452
Log	Full sample	3079	0.2389	0.0166	−0.2223	0.00205
Raw	Below 99th percentile	3048	0.0954	0.0219	−0.0735	0.01334
Log	Below 99th percentile	3048	0.2440	0.0189	−0.2251	0.00450
Raw	Below 97.5th percentile	3002	0.1188	0.0153	−0.1036	0.00163
Log	Below 97.5th percentile	3002	0.2612	0.0003	−0.2609	0.00135
Raw	Below 95th percentile	2925	0.1210	0.0209	−0.1001	0.00946
Log	Below 95th percentile	2925	0.2432	0.0002	−0.2430	0.00484

The log-HOMA coefficients are on a different exposure scale from the raw-HOMA coefficients and should not be compared numerically as if they estimated the same one-unit exposure contrast. All models retained ln(hsCRP) as the outcome and the primary period-specific adjustment structure.

Table S2: Sensitivity of the HOMA-IR-by-sex interaction to adiposity specification

Adiposity specification	<i>n</i>	Female	Male	Contrast	95% CI
No adiposity adjustment	3079	0.1892	0.0938	−0.0955	−0.1616 to −0.0293
BMI	3072	0.0606	0.0183	−0.0423	−0.0811 to −0.0035
Waist circumference (primary)	3079	0.0629	0.0082	−0.0547	−0.0914 to −0.0180
Waist-to-height ratio	3074	0.0522	0.0106	−0.0416	−0.0791 to −0.0041

Female and male columns report the estimated HOMA-IR slopes within each sex; contrast is the male-minus-female slope difference. Common-complete-case analyses among $n = 3,072$ participants produced nearly identical estimates, indicating that differences among adiposity specifications were not attributable to differential missingness. These models are sensitivity analyses of adjustment specification and are not mediation analyses.

Table S3: Targeted clinical adjustment and exclusion sensitivities

Model	<i>n</i>	Female slope	Male slope	Male–female	<i>F</i>	<i>p</i>
Smoking common-sample reference	3075	0.0628	0.0081	−0.0548	9.0765	0.00448
+ smoking status	3075	0.0632	0.0099	−0.0534	8.6687	0.00537
Statin common-sample reference	3041	0.0628	0.0080	−0.0548	9.2612	0.00412
+ statin use	3041	0.0630	0.0081	−0.0548	9.3704	0.00393
Renal common-sample reference	3075	0.0625	0.0074	−0.0551	8.9139	0.00481
+ continuous eGFR	3075	0.0623	0.0070	−0.0553	9.0634	0.00450
+ log creatinine	3075	0.0623	0.0071	−0.0552	9.0497	0.00453
ALT common-sample reference	3074	0.0625	0.0075	−0.0551	8.9101	0.00482
+ log ALT	3074	0.0633	0.0084	−0.0549	8.8098	0.00504
Alcohol common-sample reference	2919	0.0757	0.0084	−0.0673	12.3816	0.00110
+ alcohol-use category	2919	0.0750	0.0085	−0.0665	12.4028	0.00109
Exclude confirmed pregnancy	3031	0.0666	0.0099	−0.0567	9.0445	0.00454
Exclude pregnancy + cannot determine	3016	0.0668	0.0101	−0.0567	8.9438	0.00475

Each added-covariate model should be compared with its reference model fitted to the same participants. This is especially important for alcohol, for which the common-sample reference contrast differs from the full-cohort primary contrast because of sample restriction. eGFR used the 2021 CKD-EPI creatinine equation. ALT and creatinine were natural-log transformed.

Table S4: Descriptive characterization by sex and hsCRP group

Measure	Statistic	Female ≤ 10	Female > 10	Male ≤ 10	Male > 10
Unweighted n	—	1510	128	1385	56
HOMA-IR	Median	1.91	3.01	2.05	2.04
	75th percentile	2.87	4.67	3.35	4.13
	90th percentile	4.51	7.40	5.13	6.77
WBC (10^3 cells/ μ L)	Median	6.2	7.5	6.3	7.2
Absolute neutrophils (10^3 cells/ μ L)	Median	3.5	4.45	3.5	4.5
Neutrophils (%)	Median	57.4	60.0	56.6	61.4
Absolute lymphocytes (10^3 cells/ μ L)	Median	1.9	2.1	1.9	1.75
Absolute monocytes (10^3 cells/ μ L)	Median	0.5	0.5	0.5	0.7
NLR	Median	1.82	2.13	1.84	2.39

hsCRP is expressed in mg/L. CBC measures were available for all 184 participants with hsCRP > 10 mg/L. Values are descriptive and were not used to diagnose acute infection, assign an inflammatory etiology, or define additional exclusions. WBC = white blood cell count; NLR = neutrophil-to-lymphocyte ratio.